Technology and Philosophy of Artificial Intelligence

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Types of AI

- AI: software with “human-like” intelligence
- Machine Learning: AI that trains itself
- Deep Learning: ML that models neurons
- Generative AI: Deep Learning that generates text, music, art rather than just analyzing.
  - LLM Large Language Model: ChatGPT, Gemini, Claude, Grok
  - Art and Music generators: Suno, Sora, DALL-E, etc

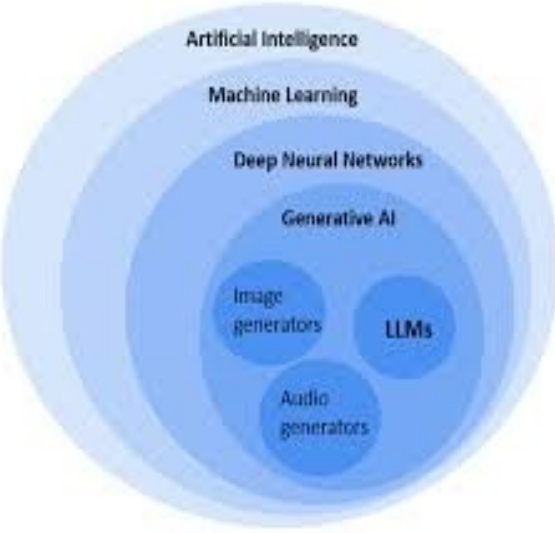


Figure 1. Illustration of LLM and AI.

LLM Demo

- Like a text conversation, but with highly sophisticated program
- Much more than a search engine:
- Write and Revise a Story (Claude Sonnet)
  - Understands the context of the story, summarizes; changes style
- Explain funny images
  - Gets the humorous intent, understands images
- Engage in “theory of mind” exercise
  - Intuitive, self-reflective

Example Uses

- Document summary/analysis – IP agreement
  - Prompt includes entire document
  - Summarizes, points out issues, correctly evaluates
  - (But check for accuracy!)
- Editing prose – remove passive voice
  - Understands grammar, style
  - Example revision for PV is correct, succinct

Example Uses, cont..

- Write standard documents – reference letter
  - Prompt should include particular details, and general instructions
  - Can reuse prompt over and over w/ adjustments
  - Good at first drafts, but needs final polish (3rd paragraph, for instance)
- Tutorial Study – hierarchical Bayes' analysis
  - Gives good "textbook" summaries and explanations
  - Also response intuitively to questions
  - Wide ranging knowledge – like talking with universal graduate TA

LLM Use “How To”

Prompting is Essential

- More than a question. Can include information, like documents, photos, etc.
- LLM “knows” info from training, sometimes years earlier. Prompt gives current details.
- Task examples – “no shot”, “one shot”, …
- Not just one prompt – entire conversation is “input” for each exchange
- Depth of remembered conversation “context window” varies by LLM
  - 10,000 words to 1,000,000

LLM “How To” – continued

- LLM’s “hallucinate”, often very confidently – check critical facts
- Many LLMs available
  - ChatGPT (OpenAI), Claude (Anthropic), Grok (X), R1 (DeepSeek)
- Steadily Improving. You can choose a level of intelligence
- Usage measured by “tokens”
  - Certain amount free. More with subscriptions
- Likely to be in short supply for a few years. Use efficiently
  - Simpler tools when they are enough
  - Ask for brief responses

An LLM Is *Not*

- An improved search engine
  - Is able to web-search when useful, but…
  - … can work unplugged from the internet
- Hand-crafted code
  - No programming specific to the questions/answers
  - Could be written in one page of code – **12 equations**
- The whole modern AI story
  - Important other areas are making major advances too, e.g. “reinforcement learning”, computer vision (will review later)

An LLM *Is*

- A sophisticated mathematical engine
  - 12 equations, but on “matrices” – huge grids of numbers
  - From 10 billion to over 1 trillion numbers
  - The “intelligence” is in those numbers (aka weights or parameters)
- Self-organizing!
  - The weights are “trained” by incremental adjustment
    - First, to match known texts
    - Then to adjust to human feedback given by “RLHF”
- Humans design the training process and the equations, but not the weights

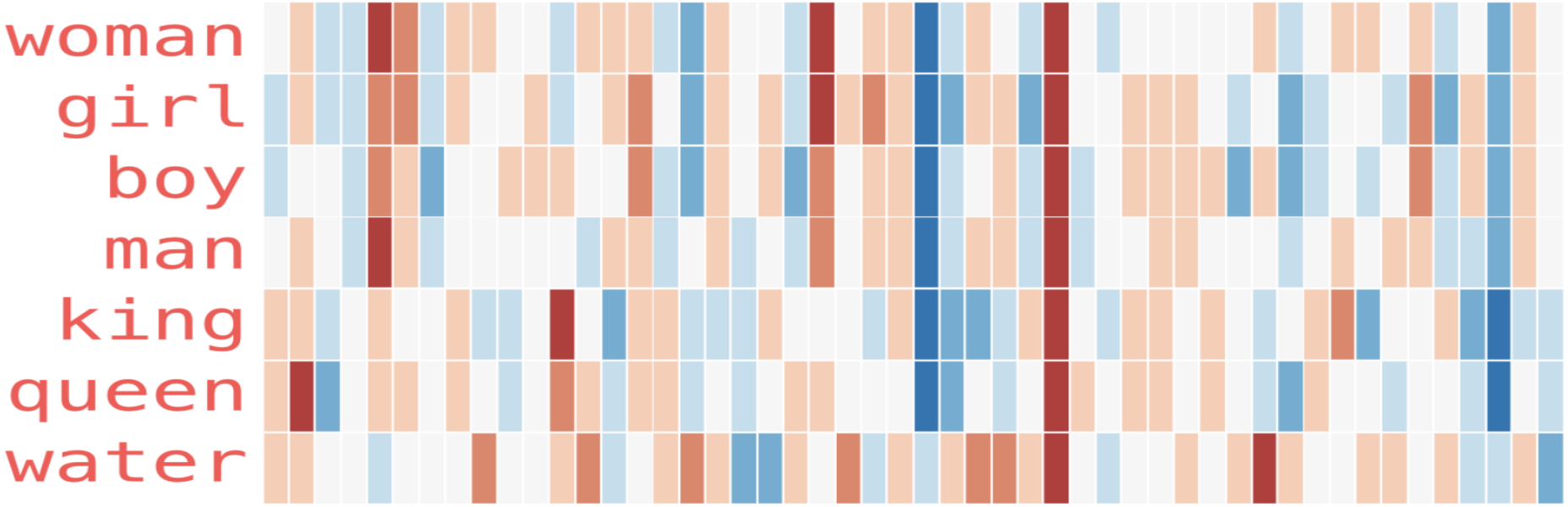
Other AI

- “Diffusion Model”
  - Deep Learning and generative, but not LLM
  - Same principles and training, but different design and purposes.
  - Generate photos, short movie clips, music
  - (Term “hallucination” originates with this type of AI)
- Can generate audio of text **in your own voice**
  - Note that this means voices are not representative e.g. on phone calls

Other AI - cont

- Computer Vision: face recognition, object tracking
- “Physical” or “Embedded” AI: robots, automated drones
- Reinforcement Learning (RL)
  - Self-instructing. Requires automatically generable feedback.
  - Examples: Alpha Go, code-writing agents, physical AI (via simulation)
- Graph Neural Networks:
  - Analyze connection patterns, e.g. chemical compounds
  - AlphaFold, which solves protein folding problems

Word/Idea “Embedding”



- This is how an LLM represents words, phrases and ideas
- Dimensions represent meaning, and also word position
- But, with 12,800 numbers per word – many conceptual dimensions

How An LLM “Thinks”

- Starts with word embeddings representing words in your prompt and the current conversation.
- Combines words via “attention blocks” into higher embeddings representing phrases, then sentences, etc.
- Predicts next “thought” as an embedding and chooses closest next word.
- Repeat, word by word...

The Weights *are* the “Intelligence”

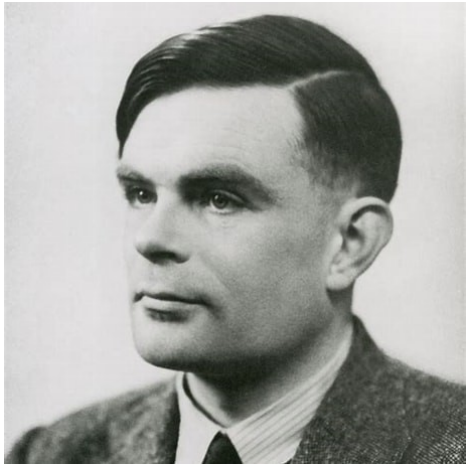
- All “reasoning” is by word/concept embeddings.
- And we (mostly) *don’t know* what each weight does.
- Could learn to converse in a language none of its programmers understand.
- Are world-models, theory of mind, etc. spontaneously emerging from the training? Latest research clearly says “yes”.

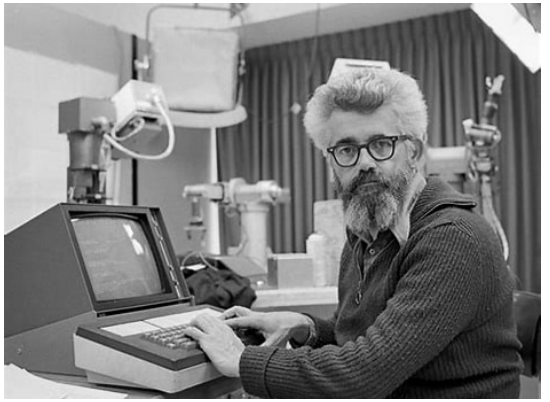
Is an LLM Intelligent?

- “Yes” arguments
  - Vast possibilities in 1 trillion weights. Printed out they’d be 4 million large books.
  - Research in “mechanistic interpretability” shows conceptual models spontaneously arise from training
  - (But, we still know only some of what those weights are doing, and “intelligent” doesn’t necessarily mean “conscious”)
- “No” arguments
  - Just mindless numbers and 12 equations
  - Just predicting the next word.
  - (But, it’s improving *fast*, a human brain is just “mindless” neurons and synapses, and *how* is it picking that next word so intelligently?)

AI Events and People 1950s

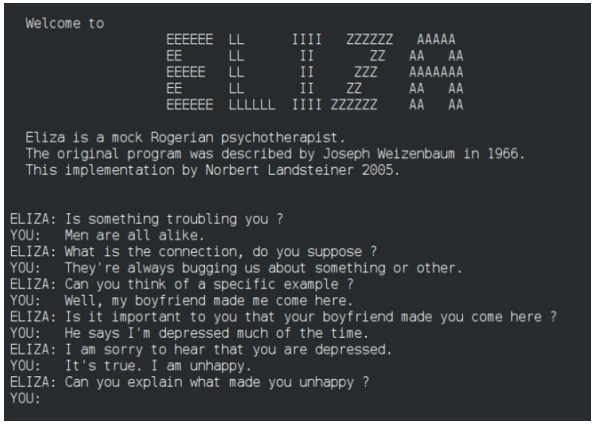
- (1912-1954) Alan Turing
  - Theoretical father of CS. “Turing Test”
  - Early thoughts on “connectionist” AI – forerunner to neural nets/deep learning
- (1956) “Dartmouth conference”
  - John McCarthy coins “Artificial Intelligence”
- (1956) Logic Theorist
  - First good “theorem prover” – symbolic AI





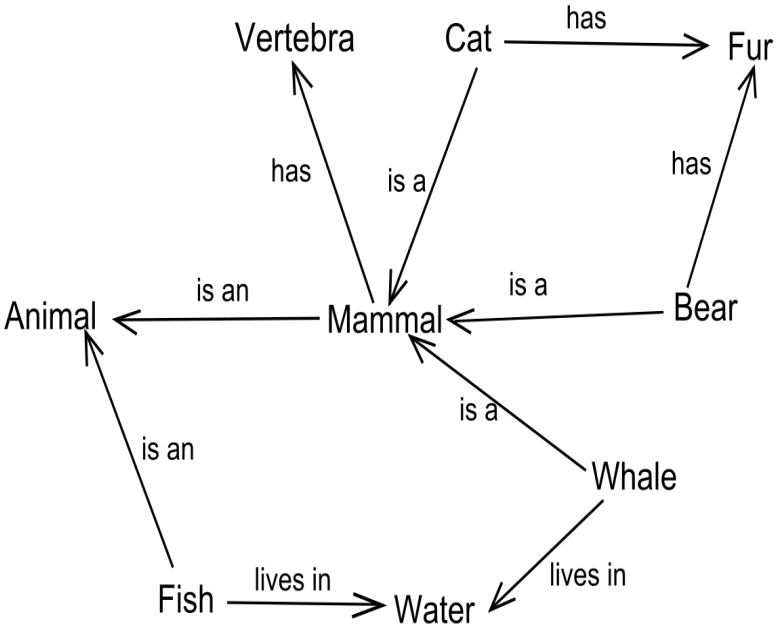
AI History 1960s

- (1965) ELIZA basic “chatbot” in 200 lines of code
- (1960s) Hopfield, Minsky and others do early work in **“neural networks”**
- (1967) Minsky: “Within a generation … the problem of creating artificial intelligence will substantially be solved.”
- (1969) Minsky and Papert publish “Perceptrons”
  - Proves one-layer (all they knew how to train) neural networks very limited and triggers first **“AI winter”**



First AI Winter 1970-1985

- (1970) S Linnainmaa quietly publishes “backpropagation” algorithm,
- (1974) P Werbos does the same. No one notices.
- Gradual development of symbol/logic based AI
- (1982) Japan announces “5th generation computing” project – hardware devoted to expert systems or symbolic reasoning
- Various *limited* successes with expert systems



AI History 1986- early1990s

- (1986) Backpropagation algorithm (re)discovered.
  - Neural nets are back.
  - But training networks > 3 layers proves very hard. (Modern deep learning uses hundreds.)
  - Advances inhibited by persistent effort to imitate neurons, which don't train by backpropagation.
- After brief burst of progress, a **second "AI Winter"**

Second AI Winter 1990-2012

- Expert systems and symbolic AI prove too limited/fragile
- Gradual improvements in neural networks, by Geoff Hinton, Ilya Sutskever, Yan LeCunn, Yoshua Bengio and many others.
- Other pattern recognition methods, e.g. SVMs, explored.
- Various limited neural net successes e.g. character recognition for written digits.

AI History 2012-2024

- Four important advances lay ground for new neural network growth
  - more available training data via internet
  - computing power via GPUs
  - better neural network design: stop imitating neurons
  - vibrant and very intensive research community
- (2012) Surprise success of neural networks in image recognition
- (2017) “Attention is all you Need” paper describes Transformers
- (2020-2023) Steady LLM improvement arrives at GPT-4

AI History – Taking Stock

- Early investigation of major ideas
  - Neural networks and symbol systems since 1950s
- Alternating optimism and AI winters
- Neural networks and ML beat symbolic logic
  - The “bitter lesson” – can’t program AI directly
- Focus on imitating neurons was a distraction
- Needed confluence of factors

2024 AI Events

- Nobel Prizes 2024
  - Physics to John Hopfield and Geoffrey Hinton
  - Chemistry to the developers of Alpha Fold
- Improved LLM Architectures
  - MoE – “Mixture of Experts”
  - CoT – “Chain of Thought” – LLMs that reason
- Early Coding Tools, e.g. Cursor

2025 Events

- Huge buildout of data centers
  - Electrical and (sometimes) water concerns
  - Shortage of computer memory
- But, ongoing efficiency advances
  - E.g. Distillation, “Turboquant”
- Mid-year talk of a plateu in LLM progress...
- But by year end, models, especially Anthropic, advancing
  - Claude Code revolutionizes software dev (!)

2026 Events

- AI Agents
  - LLM that does online tasks for you
- Ongoing relentless improvement of LLMs
- Anthropic Mythos release
  - So serious a cyber security threat that it’s being withheld

AI Events: late 2026-> (?)

- Ongoing automation of white collar work
  - Writing, coding, art, music, law, healthcare, customer service, language translation, entertainment, etc.
  - Claude Code is like a competent, knowledgeable, very fast junior engineer.
  - May increase productivity of experienced professionals by 5x,
- Physical AI
  - AI with vision, movement, hearing
  - Robotics, drones, self-driving cars, manufacturing, construction, etc.
- Scientific advances using AI

Near Term Issues to Consider

- Legal questions of copyright and liability
- “Alignment” of AI with human values
- Economic and environmental impact of data center buildout
  - Demand pressure on memory, electricity
  - Encouraging efficient use of “tokens”
- Employment Impacts
  - Not like going from farm to factory to simple white-collar work
  - “UBI” can provide subsistence, but doesn’t provide purpose
- Which nation will lead, US or China? – Manhattan project of the 2020s

Near Term Issues – Education

- How does education adapt to AI?
- Starship vs Cloister
  - Be able to do everything *without* AI.
  - Can’t be “in charge” if you are dependent
- But, do use AI
  - Leverage AI as “cointelligence” to create more than you can alone
  - Always question/challenge the AI; don’t take output at face value
  - “Any good LLM session includes you correcting the LLM”
- Students must jump the “apprentice gap”

AI Future – Limits on Prediction

- “… heavier than air flying machines are impossible”
  - Lord Kelvin, 1895
- Alchemistic Empiricism
  - Analogy of Chemistry in 1825
  - Results in fits and starts as we try things out by guess-and-by-golly
- Start with Nature, End with Design
  - Planes vs birds
  - Deep Learning vs brains : Geoff Hinton’s warning

AI Future – Ideas to Consider

- Pair of essays worth reading by Dario Amodei (darioamodei.com)
  - (2024) “Machines of Loving Grace”
  - (2026) “The Adolescence of Technology”
- Technological History Lessons
  - S curve pattern in innovation – aerospace example
  - AI winters of past
  - Effect of missing theory

AI Future – 3 Tracks

- Option A: Progress will plateau near term (year or two)
  - We are reaching limits on compute and on data
  - Example of aerospace after 1970
- Option B: Progress will plateau mid-term (2030s)
  - Continued improvement in hardware and “frontier models”
  - Likely to reach “AGI” or “superintelligence”
- Option C: Progress continues with AI building itself
  - “Singularity” concept

3 Tracks – Which?

- Estimates vary, of course, and shift over time.
  - My own vague impressions (not data supported!)
  - 2023: A: 70%, B 25%, C 5%
  - 2025 midyear A: 75%, B 15%, C 10%
  - 2026 present: A: 45%, B 40%, C 15%
- Long term, regardless of A/B possibilities
  - We have uncovered something fundamental about intelligent behavior
  - Future, near or long term, shows no clear limits

Long Term Questions

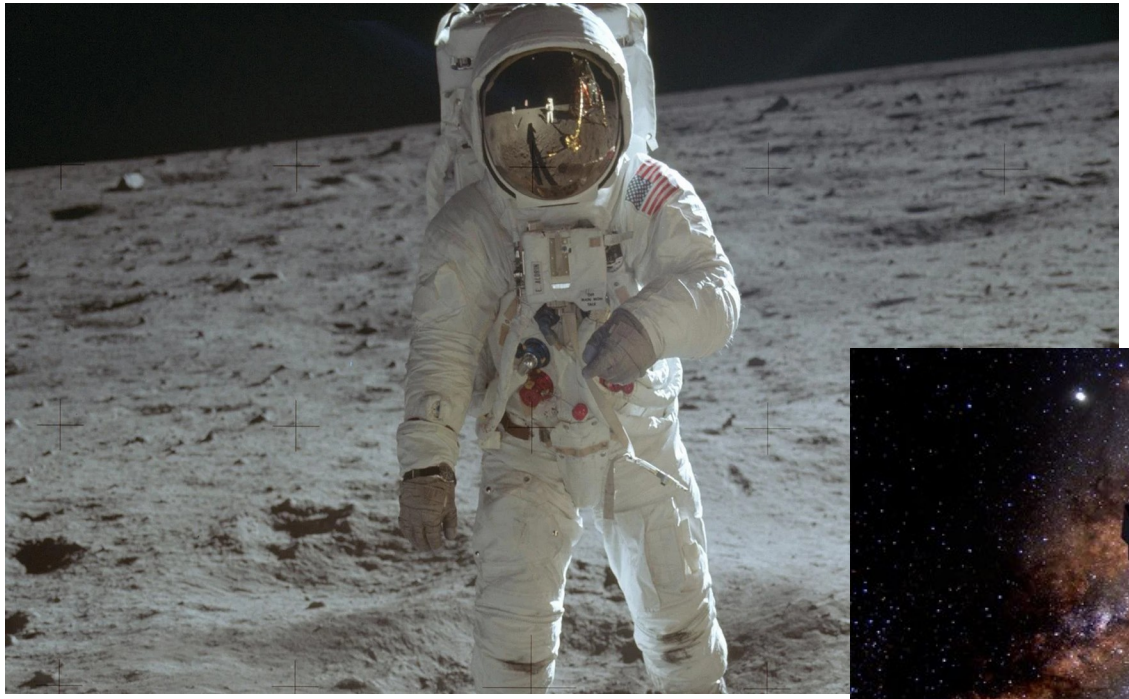
- What can AI teach us about the nature of thought and intelligence?
- What if it becomes more intelligent than us?
- Hard Problem of Consciousness
- AI as “moral patient”?
- Use of AI to extend human intelligence

Future Predictions – Dystopian SF

- Premise: Technology is a power unto itself
- (1984) Blade Runner
  - LA is stygian hellscape in 2020s
- (1984) Terminator
  - LA is nuclear-blasted wreck with robots killing humans in 2020s
- (1970) Colossus
  - AI computer becomes world dictator in late 1970s

Future Predictions – CS Perspective

- Premise: Technology is Mind's creativity unfolding.
- "The mariner will have dominion over the atmosphere and the great deep .. the astronomer will no longer look up to the stars, – he will look out from them.." Science and Health p125: 25-29







Christian Science Perspective

- *“There is no life, truth, intelligence, nor substance in matter. All is infinite Mind and its infinite manifestation…” S&H p 468*
- *“Intelligence is omniscience, omnipresence, and omnipotence. It is the primal and eternal quality of infinite Mind...” S&H p 469*